

QUESTION - 01

PRINCIPAL COMPONENT ANALYSIS :-

- PCA is the technique using which the high dimension data is represented in lower dimensions, without losing important details/information of the data.
- The algorithm gives the direction of the maximum variance through eigen value decomposition.
- Through this the image is reconstructed without losing major information.

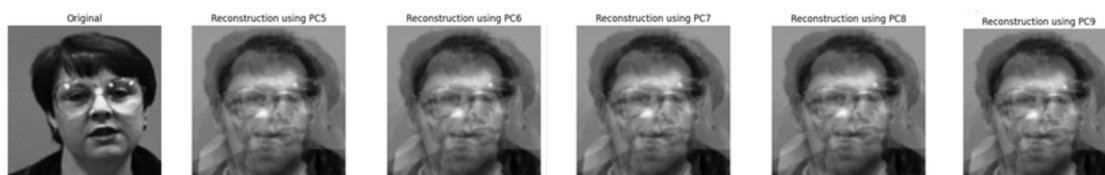
OBSERVATIONS & IMAGE RESULTS :-

- Reconstructed by projection on PC1, PC2, PC3 respectively.



Inference : Here it is clearly visible that after reconstructing the projection on the dominant eigen vector (with highest eigen value) the image is very near to the original image. As the eigen value used for reconstruction decreases the similarity with original image reduces.

- Reconstructed by projection on PC5, PC6, PC7, PC8, PC9 respectively.



Inference : It is clearly visible that using the non-dominant late eigen vector projection for reconstruction resulted in very blur and unclear images. That are tough to interpret.

Amount of Distortion is inversely proportional to the eigen value of corresponding eigen vector.

- Reconstructed by using PC1-3, PC5-9 respectively.



Inference : Here, The image is reconstructed with the help of first 3 dominant eigen vectors that are PC1-3 which resulted in almost equivalent image to original image.

Despite of using Other late eigen vector for reconstructing but the visibility of the image remain distorted.

Analysis and Findings :-

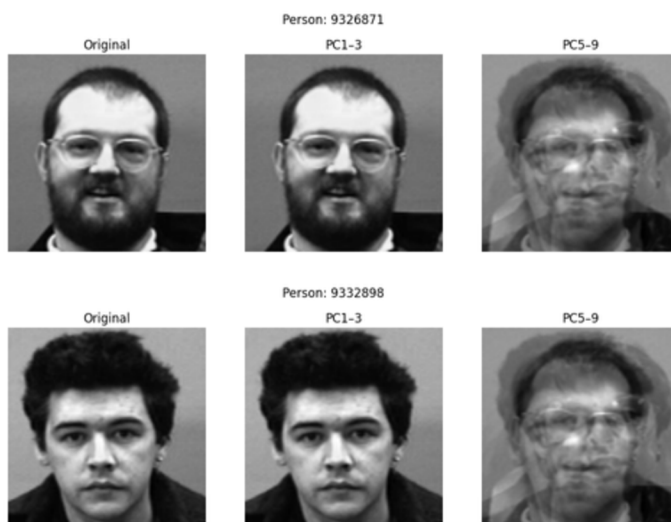
PCA was applied on grayscale, resized 100×100 face images from multiple individuals in the Face94 dataset. The main goal of the task was to achieve dimensionality reduction for higher dimension data. And understand how different component choice affects the quality and resemblance with the original image.

Effect of Varying the Number and Range of Components :-

- Very low k (e.g., $k = 3-5$):
 1. Strong dimensionality reduction and strong smoothing.
 2. captures most of the data like position of the nose, ear ,eyes and complete structure of the face.

- *Moderate k (e.g., $k = 20-50$):*
 1. It creates the good trade-off between dimensionality reduction (compression) and reconstructed image quality.
 2. Small details like hairstyle, eye color become visible.
- *High k (approaching full dimensionality):*
 1. Reconstructed image becomes indistinguishable from the original image since all the details are used to reconstruct.

Consistency Across Different Individuals and Pairs :-



Inference : For different pairs of images the results are quite consistent as visible through these reconstructed images.

For Every person the first components accounts the global structure (which captures high variance) and other are responsible for the fine detail (captures low variance) addition.

Key Findings and Insights :-

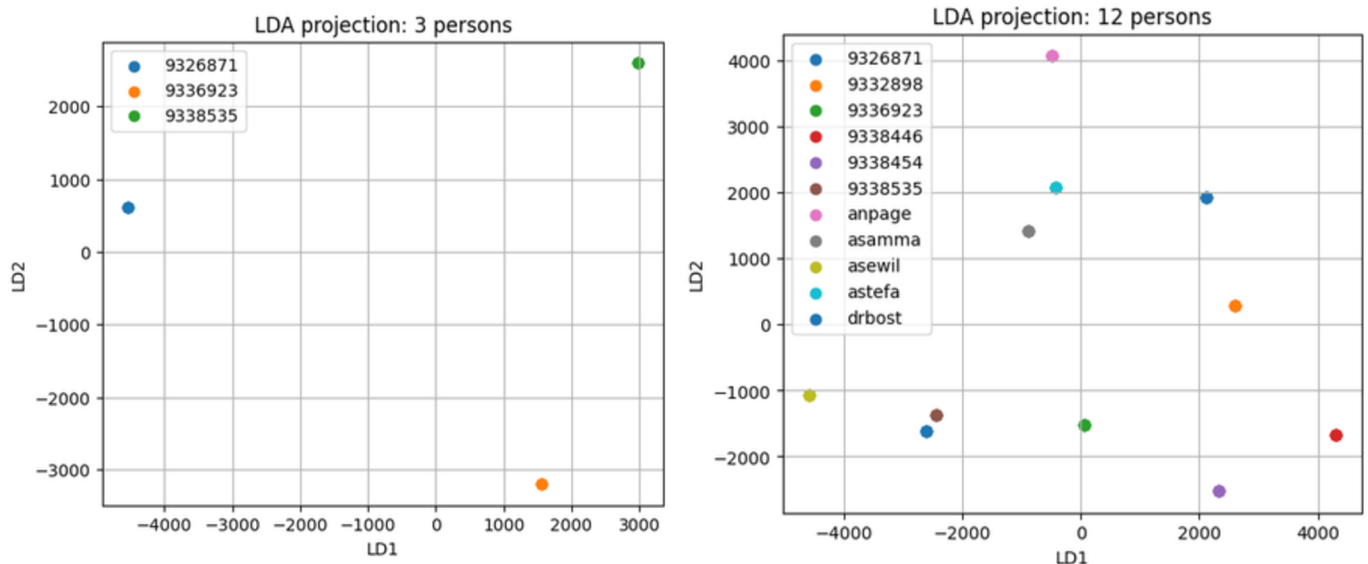
- PCA as Global Structure separator :-
 1. PCA algorithm gives the direction of the maximum variance and that direction accounts for the global structure in any face image.
- Trade-off Between Fidelity and Dimensionality reduction :-
 1. As per the requirements of the details in the image one may use some first dominant eigen vector for reconstruction and achieve quality and compression at the same time.

QUESTION - 02

LINEAR DISCRIMINANT ANALYSIS :-

- LDA is the unsupervised Machine learning technique used for dimensionality reduction and classification tasks.
- Goal is to find to find linear combination of the features that maximizes separation between the classes and minimizes spread in the class.
- Maximizing the Mean difference.
- Minimizing the variance within the class.

OBSERVATIONS & ANALYSIS :-



INFERENCE :-

- LDA was then used to project the high-dimensional data into a 2-dimensional discriminant space (LD1-LD2).
- In the scatter plot samples from the each person form tight cluster, while the different individual appear clearly separated at distinct positions.

ANALYSIS :-

LDA gives the linear combination of the feature such that it maximizes the separation between the two person images and minimizes their variance.

This is also clearly visible in the plot where the 2 discriminant dimensions are used to visualize and for every person it distinct position in the plot.

Discussion of findings and insights

- LDA is the method which is helpful in classification related tasks, since it captures the data point according to the max mean difference and minimum spread in individual class.
- The process of LDA is computationally heavy and it become extensive when dealing with large number of dataset, it required high time to provide the result. This is mainly due to the eigen vector and value calculation involved in the process that too for high row and column matrix.

DIFFERENT CHOICES OF PARAMETERS :-

- Number of images : When the three images were given the plot accounts for the highly separable characteristics that means the images are of different person. On increasing the number of person the plot became more scattered but the classification was possible.
- Another parameter choice is the number of discriminant dimensions. Here the two directions LD1 and LD2 were used and the three person occupied different regions.

COMPARISON OF LDA AND PCA :-

- PCA: Unsupervised; finds directions of maximum overall variance, it is more dependent on the global structure that may not be the classification parameter, so different people can overlap in low-D plots.
- LDA: Supervised; uses labels to maximize between-class separation while minimizing within-class spread, so same-person samples cluster tightly and different persons separate clearly.
- For distinguishing individuals, LDA is preferred because its objective is class separation, whereas PCA is better for compression/reconstruction.

QUESTION - 03

SINGULAR VALUE DECOMPOSITION :-

- Singular Value Decomposition is also a Machine Learning techniques used in dimensionality reduction. High dimension data is converted to lowered ones.
- For this reduction it uses the technique of decomposition, where the matrix is fragmented into the three matrices U , σ , V^T
- After this, we try to reconstruct the image using the first k eigen vectors that capture the most of the information.

IMAGE RESULTS (original, noisy, reconstructed) :-



Observations and analysis (varying k (no. of eigen vectors) :-

- As soon as the noise is added to the image, distortions are visible.
- After SVD the Image obtain is less noisy than the previous noisy image.
- For $k=5$:- $k=5$, the reconstruction is very smooth.
- For $k=20$:- $k=20$, the reconstruction recovers much more facial detail and decreasing the large percentage of the noise, also the person is quite recognizable.
- For $k=50$:- $k=50$, the reconstruction becomes close to the noisy image: fine textures and sharp edges are clearer, but as the number of eigen vector used for construction imposed back noise to the image.

HOW SVD HELPS IN REDUCING NOISE :-

- SVD expresses the image as $A=U\Sigma V^T$, where large singular values correspond to dominant, smooth structures (overall intensity and global facial shape), and small singular values capture finer, high-frequency variations.
- Random noise mainly affects these small-energy components, so by truncating the decomposition and keeping only the top k singular values, many noise-dominated components are discarded.

DISCUSSION OF FINDINGS AND INSIGHTS :-

Using a small k yields strong noise removal but produces overly smooth. Using a large k preserves fine details and texture but also retains original noise. An intermediate k (around 20 in this experiment) offers the best outcome.